

41) Use quantifiers to express the associative law for multiplication of real numbers.

$$\forall x \forall y \forall z ((x \cdot y) \cdot z = x \cdot (y \cdot z)) \text{ Common domain: Set of all real no.s}$$

42) Use quantifiers to express the distributive laws of multiplication over addition of real numbers.

$$\forall x \forall y \forall z ((x+y) \cdot z = x \cdot z + y \cdot z) \text{ Common domain: Set of all reals}$$

$$\forall x \forall y \forall z (x \cdot (y+z) = x \cdot y + x \cdot z)$$

43) Use quantifiers and logical connectives to express the fact that every linear polynomial (that is, polynomial of degree 1) with real coefficients and where the coefficients of x is non-zero, has exactly one real root.

$$\forall a \forall b ((ax+b=0) \wedge (a \neq 0) \Rightarrow x = -(b/a)) \rightarrow \text{This is wrong}$$

$$\forall a \forall b ((a \neq 0) \wedge \exists x (ax+b=0) \wedge \forall w ((aw+b=0) \Rightarrow (w=x))) \rightarrow \text{Right}$$

Common domain: Set of all real no.s

44) Use quantifiers and logical connectives to express the fact that a quadratic polynomial with real number coefficients has at most two real roots.

$$\forall a \forall b \forall c ((a \neq 0) \wedge \exists x \exists y ((ax^2+bx+c=0) \wedge (ay^2+by+c=0) \wedge \forall w ((aw^2+bw+c=0) \Rightarrow (w=x) \vee (w=y)))) \text{ Common domain: Set of all real numbers.}$$

47) Show that the two statements $\neg \exists x \forall y P(x,y)$ and $\forall x \exists y \neg P(x,y)$, where both quantifiers over the first variable in $P(x,y)$ have the same domain, and both quantifiers over the second variable in $P(x,y)$ have the same domain, are logically equivalent.

$$\neg \exists x \forall y P(x,y) \equiv \forall x \neg \forall y P(x,y) \text{ (De Morgan's law of quantifiers)}$$

$$\equiv \forall x \exists y \neg P(x,y) \text{ (De Morgan's law of quantifiers)}$$

Note: This logical equivalence is only possible due to the given constraint on domains.